Homework 5

Combinatorics

Fall 2026

Due: Thursday, October 1, 2026 23:59
Submit: Gradescope, through Blackboard's assignment area.

Please write your homework by hand. Start each exercise on a new page, scan your work to a legible PDF, and clearly mark the pages associated with every exercise. For the computational parts, you may include a printout or screenshot of your Sage code and output.

1. (A constrained sum)

Points: 14

Let a_k be the number of ordered triples (u, v, w) of integers satisfying

$$u + v + w = k,$$

where u is nonnegative and even, v is positive and odd, and $10 \leq w \leq 20$.

(A) **6 points.** Let $U(x)$, $V(x)$, and $W(x)$ be the generating functions for the allowed choices of u, v, and w, respectively. Write each generating function both as a series and in closed form.

(B) **2 points.** Then find the generating function for the required problem in closed form.

(C) **6 points.** Use Sage to find the number of solutions for $k = 20$, $k = 35$, and $k = 50$. Include your Sage code and output with your work (screenshot is fine)

Need a free hint?. For part (C), begin with `var('x')`, define A using the closed form from part (A) and (B), and use `A.series(variable, order)` before extracting the three coefficients. Extract coefficients with `A.series(variable,order).coefficient(variable,power)`

2. (Computing a closed formula for a sum of cubes)

Points: 16

In this exercise we use generating functions to compute a closed formula for the sum of the cubes of numbers j from 0 to k.

Follow the steps below and show the computations used in each step.

(A) **2 points.** Write down the generating function in closed form for the sequence

$$1, 1, 1, 1, 1, 1, 1, ...$$

(B) **2 points.** Find the generating function in closed form for the sequence

$$0, 1, 2, 3, 4, 5, 6, ...$$

(C) **2 points.** Find the generating function in closed form for the sequence

$$0, 1 \cdot 1 \cdot 1, 2 \cdot 2 \cdot 2, ...$$

(D) **3 points.** Find the generating function in closed form for the sum for cubes up to k.

(E) **6 points.** Find

$$[x^m] \frac{1}{(1-x)^5}$$

and use that to extract the coefficient of x^k from $Q(x)$.

3. (Compositions with parts 1, 2, and 3)

Points: 12

Let r_k be the number of ordered compositions of k whose parts all belong to $\{1, 2, 3\}$. Include one empty composition when $k = 0$. For example, $(1, 3)$ and $(3, 1)$ are different compositions of 4.

(A) **3 points.** Write the generating function $M(x)$ for one part, and explain why $M(x)^m$ counts compositions having exactly m parts.

(B) **4 points.** Sum over all possible numbers of parts to prove that

$$R(x) = \sum_{k \geq 0} r_k x^k = \sum_{m \geq 0} M(x)^m = \frac{1}{1 - x - x^2 - x^3}.$$

(C) **5 points.** Use Sage to find r_{30}. Also report the list $(r_0, r_1, \ldots, r_{12})$. Include your Sage code and output with your work.

You may use https://sagecell.sagemath.org/. The following sample shows how to extract coefficients from a power series; adapt it to this problem.

```
var('x')
F = 1 / (1 - x - x^2)
T = F.taylor(x, 0, 30)
print(T.coefficient(x, 20))
print([T.coefficient(x, k) for k in range(11)])
```

Need a free hint?. For one part, the exponent records its value. A composition has a unique number m of parts, so summing the powers $M(x)^m$ neither misses nor repeats any composition.

4. (Ordered compositions versus restricted partitions)

Points: 10

Let p_k be the number of ways to write k as a sum of parts chosen from

$$\{1, 3, 4, 6\},$$

where each part size may be used any number of times and order is irrelevant. Thus $1 + 3 + 3$ and $3 + 1 + 3$ represent the same choice of parts.

(A) **5 points.** Model $(p_k)_{k\geq 0}$ with an ordinary generating function $P(x)$. Write and explain one factor for each allowed part size. In particular, explain why this model is a product of factors rather than the sequence construction used in Exercise 3.

(B) **5 points.** Use Sage to compute p_{25}, p_{50}, and p_{100}. Include your Sage code and output with your work. Make sure your Taylor expansion includes the coefficient of x^{100}.